

MathFoundations237:

**Introducing (finally!)
Box Arithmetic**

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MF237: Introducing (finally) Box Arithmetic

Zero: $\boxed{} = 0$

①

Nat

zero's

$$\boxed{\boxed{}\boxed{}} = 2$$

Poly

nat's

$$\boxed{\boxed{\boxed{}\boxed{}}\boxed{}} = 1 + \alpha^2$$

Mult

poly's

$$\boxed{\boxed{\boxed{\boxed{}\boxed{}}\boxed{}}\boxed{\boxed{}\boxed{}}} = 2 + \alpha_0 + \alpha_0 \alpha_2$$

Oper

Addition:

$$\boxed{\boxed{}\boxed{}} + \boxed{\boxed{}} = \boxed{\boxed{\boxed{}\boxed{}}\boxed{}\boxed{}}$$

$b: a \in A, b \in B$

$c, d \in D \dots$

atives !!

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Zero: $\square = 0$

①

Nat: mset of zero's $\boxed{\square\square} = 2$

Poly: mset of nat's $\boxed{\boxed{\square\square}\square} = 1 + \alpha^2$

Multi: mset of poly's $\boxed{\boxed{\square\square}\square}\boxed{\square\square} = 2 + \alpha_0 + \alpha_0\alpha_2$

Operations: addition: $\boxed{\square\square} + \boxed{\square} = \boxed{\square\square\square}$

multiplication: $A \times B \equiv [a + b : a \in A, b \in B]$

ret: $C \wedge D \equiv [c \times d : c \in C, d \in D] \dots$

when we introduced... negatives !!

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Zero: $\square = 0$

1 set of zero's $\square\square = 2$ $1 + \alpha_0^2$

1 set of nat's $\square\square\square = 1 + \alpha^2$

poly's $\square\square\square\square = 2 + \alpha_0 + \alpha_0\alpha_2$

tion: $\square\square + \square = \square\square\square$

$[a + b : a \in A, b \in B]$

$c \in C, d \in D]$

negatives

...
 $\begin{matrix} p_0 \\ \downarrow \\ p_0^* \end{matrix}$

Box Arithmetic

we want to go back, and introduce
 +s and negatives as an essential
 from the start!

a Box



X^a



try

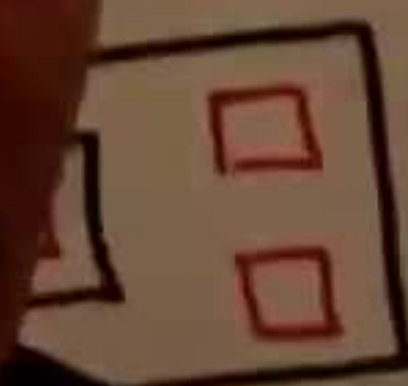
MF237: Introducing (finally) Box Arithmetic

Zero: $\square = \cap$

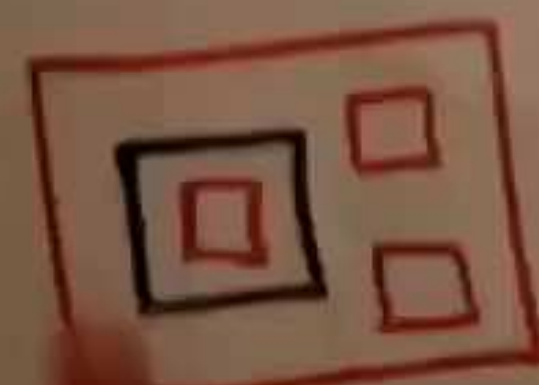
Box Arithmetic

... want to go back, and introduce
... and negatives as an essential
... from the start!

Box



a Box



$\times^a$



metry

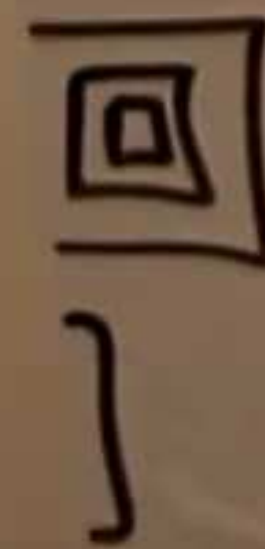
②

①

$$1 + \alpha_0^2$$

$$\alpha^2$$

$$1_0 + \alpha_0 \alpha_2$$



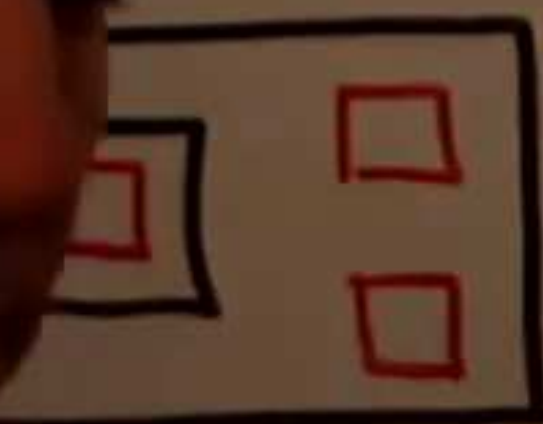
$\therefore$
 $\rightarrow$
 $\times$

Box Arithmetic

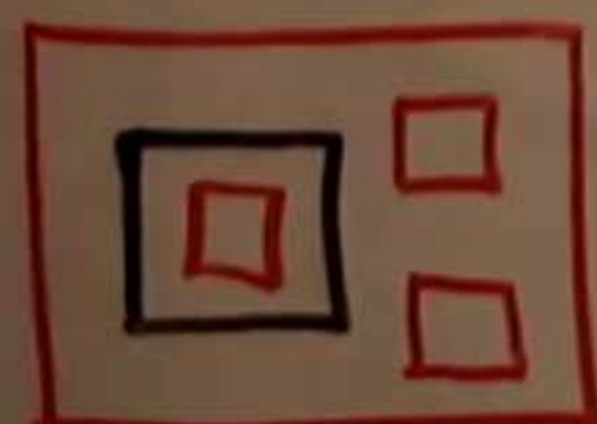
②

We now want to go back, and introduce
sets and negatives as an essential
from the start!

Box



a Box



$\times^a$



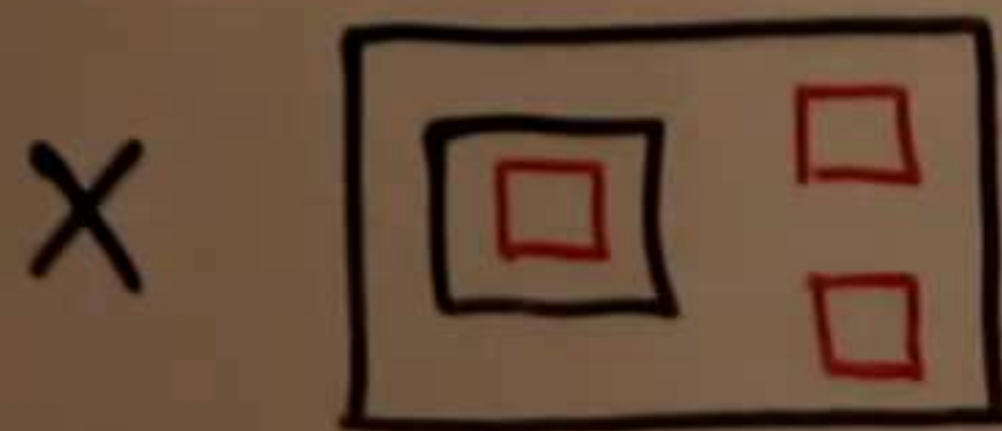
x Symmetry

Box Arithmetic

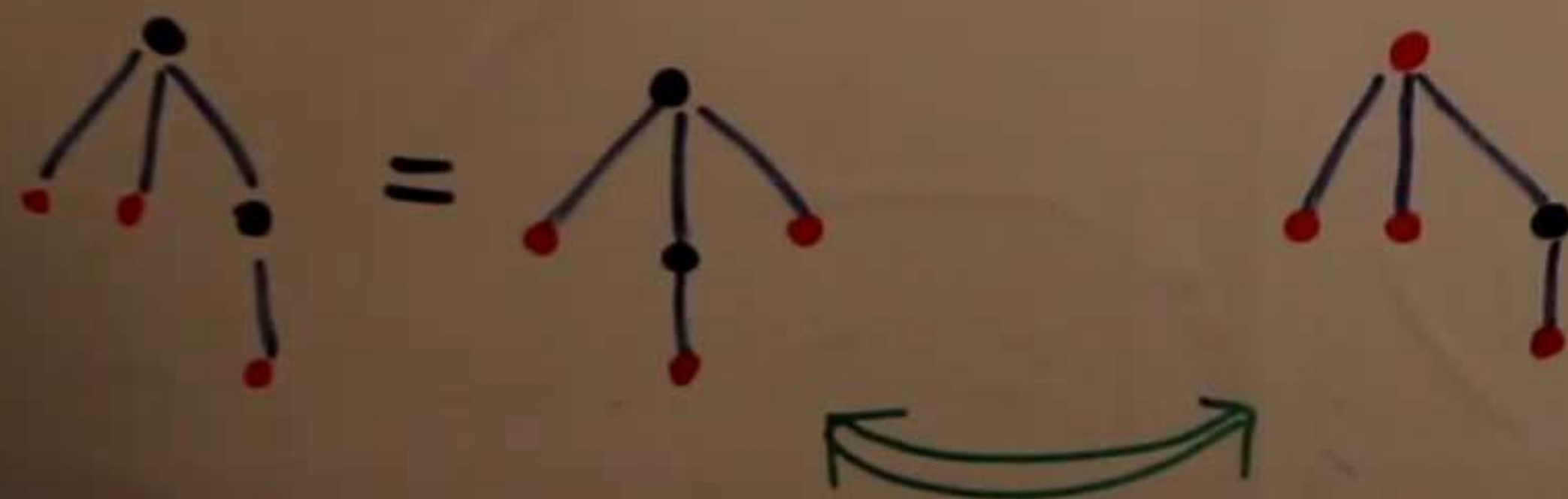
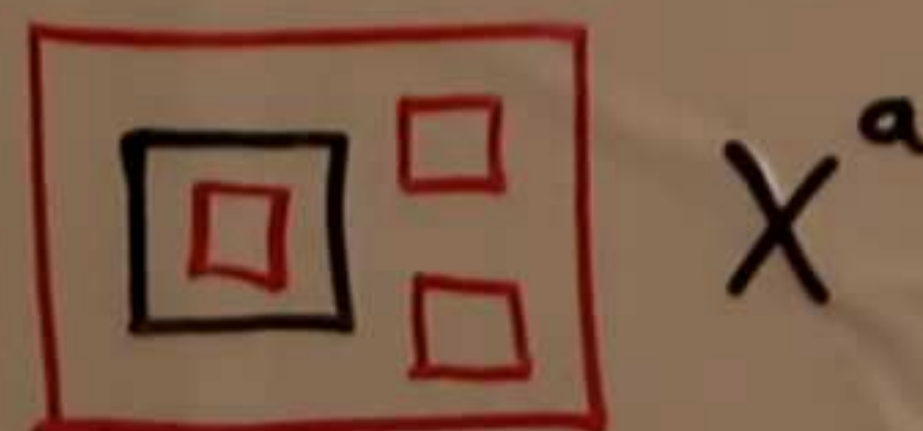
②

We now want to go back, and introduce anti-msets and negatives as an essential feature from the start!

Box



aBox



Box/aBox symmetry

Box Arithmetic

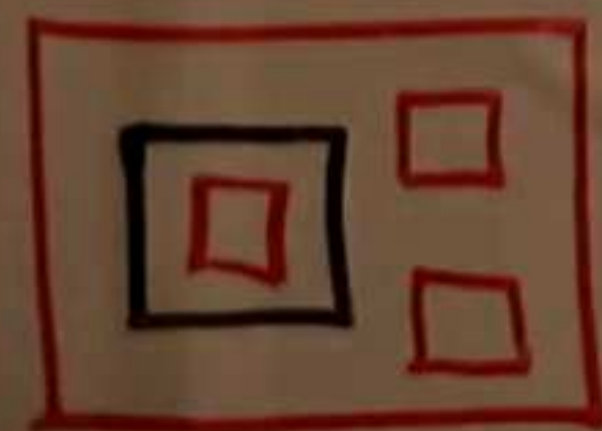
②

We now want to go back, and introduce anti-r and negatives as an essential feature to the start!

$-2 + \alpha$

aBox $= (-2 + \alpha^{-1})^a$

X



$\times^a$

Box/aBox
creation/
annihilation

$\boxed{X X^a} = \square$

$\boxed{X X^a} = \square$

etry

Next in the series:

MathFoundations238:

**Box Arithmetic with
polynumbers**

